
A. BUILDING OVERVIEW

Occupancy Type : R-2 Residential (upper), B Business (mid), M Mercantile (podium)

Construction Type : Type I-A (non-combustible)

Stories Above Grade: 42 + 2 mechanical penthouses

Stories Below Grade: 3 (parking + MEP plant)

Building Height : 518'-6" to top of parapet

Footprint : 185' x 120' (22,200 SF typical floor plate)

B. STRUCTURAL SYSTEM

Foundation:

- 84 drilled caissons, 48" dia., rock socketed min 10 feet into dolomite bedrock
- Caisson cap grade beams: 4000 psi concrete
- Mat slab at B3 level: 36" thick, 5000 psi

Superstructure:

- Steel columns: W14x342 to W10x49 (ASTM A992)
- Steel beams: W24x84 to W16x40; composite deck 3" LW concrete on 2" metal deck
- Lateral system: core shear walls (24" RC walls) + perimeter moment frames
- Estimated structural steel: 14,200 tons

Floor-to-Floor Heights:

- Podium (L1?L6): 14'-0" typical
- Office (L7?L28): 13'-6" typical
- Residential (L29?L42): 10'-0" typical

C. ENVELOPE & FACADE

Curtain Wall : Unitized aluminum system, 285,000 SF glazed area

Glazing : Low-iron insulated glass, U-value 0.28, SHGC 0.25

Waterproofing : Self-adhered membrane + fluid-applied at podium levels

Roofing : TPO 80-mil membrane on tapered insulation (penthouse roofs)

D. MEP CAPACITY (DESIGN LOADS)

Electrical Service : 12,000 A, 480/277V, 3-phase, 4-wire

Emergency Generator: 2,500 kW diesel (life safety + elevators)

HVAC : Four-pipe fan coil (residential); VAV (office)

Plumbing Fixtures : 1,840 total (residential + common + retail)

Fire Protection : Wet sprinkler throughout; standpipes Stages 1?4

Elevators : 18 passenger + 2 freight (Otis Gen3, destination dispatch)

E. MATERIAL QUANTITIES (ESTIMATED FROM TAKEOFF)

Material	Quantity	Unit
Ready-Mix Concrete (all strengths)	68,500	CY
Structural Steel (fabricated)	14,200	TON
Rebar (#4 through #11)	2,850	TON
Metal Deck (1.5" composite)	890,000	SF
Curtain Wall Panels	4,200	EA
Fire-Rated Drywall (Type X)	1,450,000	SF
Copper Pipe (Type L)	42,000	LF
EMT Conduit (1/2" ? 4")	185,000	LF

F. INSPECTION HOLD POINTS (STRUCTURAL)

1. Caisson bearing verification (each shaft)
2. Rebar pre-pour (foundations, shear walls, decks)
3. Structural steel bolted connection inspection (10% sample)
4. Structural steel welded connection inspection (100% critical)
5. Composite deck attachment verification
6. Fireproofing thickness survey (spray-applied)

7. Curtain wall water test (ASTM E1105 mock-up + 5% field)

G. PHASED CONSTRUCTION SEQUENCE (FROM SCHEDULE NARRATIVE)

Phase 1 ? Mobilization & Demolition	Month 1?2
Phase 2 ? Excavation & Foundation	Month 2?7
Phase 3 ? Structural Steel Erection	Month 6?14
Phase 4 ? Building Enclosure	Month 12?20
Phase 5 ? MEP Rough-In	Month 14?24
Phase 6 ? Interior Fit-Out	Month 20?27
Phase 7 ? Commissioning & Closeout	Month 26?28

END OF BLUEPRINT SUMMARY ? FOR CONSTRUCTAIQ AGENT TESTING